

Original papers

SLAM-PYE: Tightly coupled GNSS-binocular-inertial fusion for pitaya positioning, counting, and yield estimation

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ABSTRACT

Accurate pre-harvest yield estimation facilitates the more rational allocation of resources. Previous methods of estimating fruit position, either directly from hardware or through Structure from Motion (SfM), encounter challenges such as high hardware costs, extensive computational resources, and repetitive counting. In this paper, we present a graph optimization-based system that tightly couples dual-frequency GNSS raw measurements and visual-inertial data to estimate pitaya state, including its position and radius. Firstly, the system utilizes a coarse-to-fine approach to estimate the transformations on three axes and align three complementary sensors to a unified coordinate system, thus reducing the computational complexity of pure visual navigation. Secondly, for the first time, this system analytically calculates the pitaya state using the single-view measurement output from the binocular camera, replacing expensive lidar. It employs this calculated state as the initial value, then optimizes the pitaya state and eliminates outliers by multiple reprojection residuals from multi-view measurements. Next, for the first time, we integrate ionospheric-free pseudorange and Doppler residuals into the visual-inertial factor graph, thereby mitigating multipath interference in double-layer film greenhouse and providing a unique ID and position for each pitaya. Finally, we calculate the weight of each pitaya using a cubic polynomial based on its radius and estimate the overall greenhouse yield by summing all individual weights. In greenhouse experiments, our system achieved an Root Mean Square Error (RMSE) of 7 cm for position estimation, 7 mm for radius estimation, and a weight Mean Absolute Percentage Error (MAPE) of about 16% for individual pitaya. Overall, our system can provide cost-effective, real-time, non-repetitive, and low-missing pitaya yield estimation to assist greenhouse managers in decision-making.

1. Introduction

Cheap, fast, simple, and precise yield estimation is desired by agricultural practitioners. It plays a vital role in risk management throughout various stages of fruit production, including picking, warehousing, transportation, and sales. The traditional method of estimating orchard yield, which involves weighing fruits at multiple sampling points, faces challenges related to sampling quantity and distribution strategies (Wulfschuh et al., 2012). Subsequently, prior studies have proposed numerous automatic yield estimation systems, categorized as direct or indirect methods based on whether the measurement is specific to the individual fruit (He et al., 2022).

The indirect method assesses orchard yield by constructing a statistical model that incorporates various yield-related factors, including meteorological conditions, geographical conditions, management strategies, and plant characteristics. For instance, Obsie et al. (2020)

utilized the composition of bee species and meteorological conditions to predict blueberry yields, while Bai et al. (2021) employed multispectral images and solar radiation data to predict apple yields. Similarly, Matese and Di Gennaro (2021) used multispectral images, meteorological conditions, and canopy volume to estimate grape yields. Although the indirect method does not directly measure specific fruits, it has the ability to provide a broad range of macroscopic estimates. However, assembling a comprehensive dataset of factors poses challenges due to the multifaceted nature of yield determination. Furthermore, unique regional characteristics make generalizing statistical models across different areas challenging.

Conversely, the direct method focuses on determining the size of each fruit, and then calculating the total yield of the orchard by aggregating individual weights (Miranda et al., 2023b). Dorj et al. (2017) utilized traditional image processing techniques for citrus detection, whereas Santos et al. (2020) employed various neural networks

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for grape detection and tracking. Li et al. (2023) enhanced YOLOv5 for tomato detection and position estimation using a depth camera, although this hardware may suffer from instability under intense sunlight. To address this issue, Hou et al. (2022) used the kriging method to recover missing depth readings for citrus in case of camera measurement failure. In contrast, Kang et al. (2022) opted to utilize sunlight-insensitive lidar to assist the depth camera in achieving robust and accurate positioning of apples.

Despite the evident limitations of the depth camera, it remains widely employed in radius estimation and fruit reconstruction tasks owing to its rich data and moderate cost. Yu et al. (2022) used a depth camera to measure the radius of pomegranate, achieving an error of 2.4 mm. Gong et al. (2022) employed a depth camera to attain a center positioning accuracy of 0.6 mm for partially occluded tomatoes. Additionally, they conducted an analysis of the correlation between lighting conditions and accuracy of depth camera. Miranda et al. (2023a) utilized a different depth camera that is more resilient to sunlight, to measure the radius of an unoccluded apple, achieving in a weight error of 5% by allometric modeling. To mitigate the impact of sunlight variations on measurements, Liu et al. (2023) integrated a depth camera and lidar, achieving in a position error of 21 mm for the apple in outdoor experiments. In terms of orchard reconstruction, Chen et al. (2021) opted to integrate a binocular camera and a robotic arm for reconstructing a dense map. Xiong et al. (2023) utilized OctoMap for optimizing the dense map, achieving an average memory saving rate of 97%.

Additionally, apart from utilizing 3D measurement directly from hardware, software triangulation can recover 3D information lost in the projection. For example, Rakun et al. (2019) proposed a method to estimate the radius and distance of plants or fruits through image registration and template matching. They introduce geometric principles and compare various pixel matching methods. In addition, Herrero-Huerta et al. (2015) used camera photos to reconstruct a 3D model of the grape bunch, achieving a coefficient of determination of 0.78 between the estimated and actual weights. Luo et al. (2016) estimated the cutting point and volume of grape bunches through binocular matching and curve fitting. Operating at a speed of 0.7 seconds per frame, the errors for both height and diameter are less than 20 mm. Furthermore, Gené-Mola et al. (2021) employed Structure from Motion (SfM) and multi-view binocular technology to reconstruct complex point clouds over a wider range, utilizing multiple models to estimate the radius and visualize the fruit. This study confirmed that relying solely on vision enables high-precision radius estimation, with an error of approximately 4 mm and taking 46 min.

Despite the compromises on equipment cost and computing resources, the aforementioned methods successfully achieve accurate estimation of fruit state within a limited region. If the device's motion trajectory involves multiple passes forth and back, or if the fruit is occluded during single-view measurement, and the system lacks sensors for global position measurement, it may result in repetitive counting. Experiments conducted by Rapado-Rincón et al. (2023) utilize a robotic arm and lidar to track and count tomatoes. The results demonstrate that the multi-view measurement can effectively reduce counting errors caused by occlusion and ID switching. In addition, the integration of Global Navigation Satellite System (GNSS) measurements into the system ensures the uniqueness of the position in the map. However, multipath interference in the greenhouse makes fixing the solution challenging for Real-Time Kinematic positioning (RTK) engines and introduces jumps in the outcomes of Single-Point Positioning (SPP) algorithms (Pini et al., 2020). Mitigating multipath interference by linearly combining multiple raw GNSS measurements at various frequencies has been demonstrated to be effective (Kamatham and Vemuri, 2017). Moreover, the introduction of inertial data into the system ensures continuity between successive frames (Mur-Artal and Tardós, 2017b).

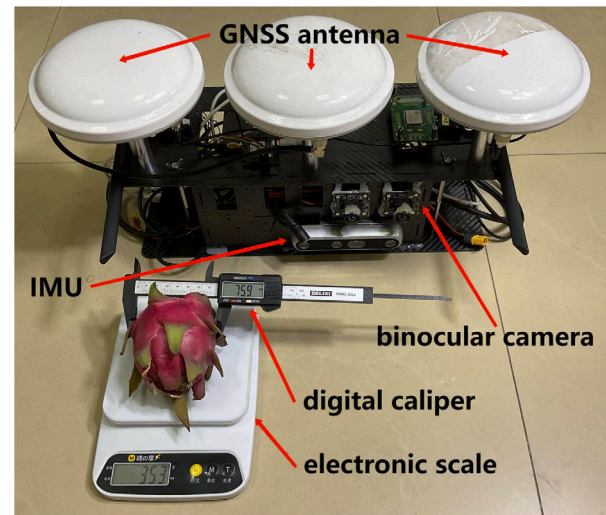


Fig. 1. The measuring equipment utilized in the experiments. The digital caliper has an accuracy of 0.1 mm. The electronic scale has an accuracy of 1 g. A homemade binocular camera with a baseline of 55 mm, a depth camera with a built-in IMU, three GNSS antennas, and two GNSS receivers are affixed to the handheld platform. The image and depth output of the Intel RealSense D455 depth camera is blocked, and solely inertial data is output. In the arrangement from left to right, the three antennas are the UM982 secondary antenna, the ZED-F9P primary antenna, and the UM982 primary antenna. Detailed specifications are presented in Table 2.

Overall, the previous systems for estimating fruit yield have several limitations. Firstly, sensors capable of directly measuring the state of fruit are often expensive, while our system achieves this function by a combination of cost-effective sensors. Secondly, SfM can restore fine but unnecessary textures, while our system reconstructs only sparse mappoints and pitaya for real-time performance. Additionally, sensors lacking the capability of global position measurement may present varying relative positions for the same fruit, while our system integrates GNSS raw measurements, ensuring the unique position of each pitaya and avoiding repetitive counting. Lastly, the fixed perspective causes certain fruits to be obscured and ignored, whereas our system measures the pitaya radius from multiple perspectives, facilitating the exclusion of outliers and enhancing the accuracy. Therefore, the objective of our system is to tightly couple complementary sensors to obtain a unique and accurate estimation of position and radius, and ultimately estimate the yield of the entire greenhouse inexpensively and rapidly based on pitaya's radius and quantity. The main contributions of this article are as follows.

- A graph optimization-based approach that tightly couples Visual-Inertial (VI) data and dual-frequency GNSS raw measurements. It integrates ionospheric-free pseudorange and Doppler residuals to the VI factor graph for the first time.
- An innovative analytical algorithm for calculating pitaya position and radius from single-view measurement, as well as their reprojection residuals from multi-view measurements.
- An empirical cubic polynomial with respect to radius for estimating the weight of individual pitaya.
- An evaluation of the proposed system regarding its stability, accuracy, and real-time performance in both indoor and greenhouse experiments.

2. System overview

In terms of hardware, the system incorporates three types of heterogeneous sensors. Given that pitaya typically has a ground clearance of less than one meter, roughly equivalent to the height of a standing person's elbow, the sensors are mounted on a handheld platform,

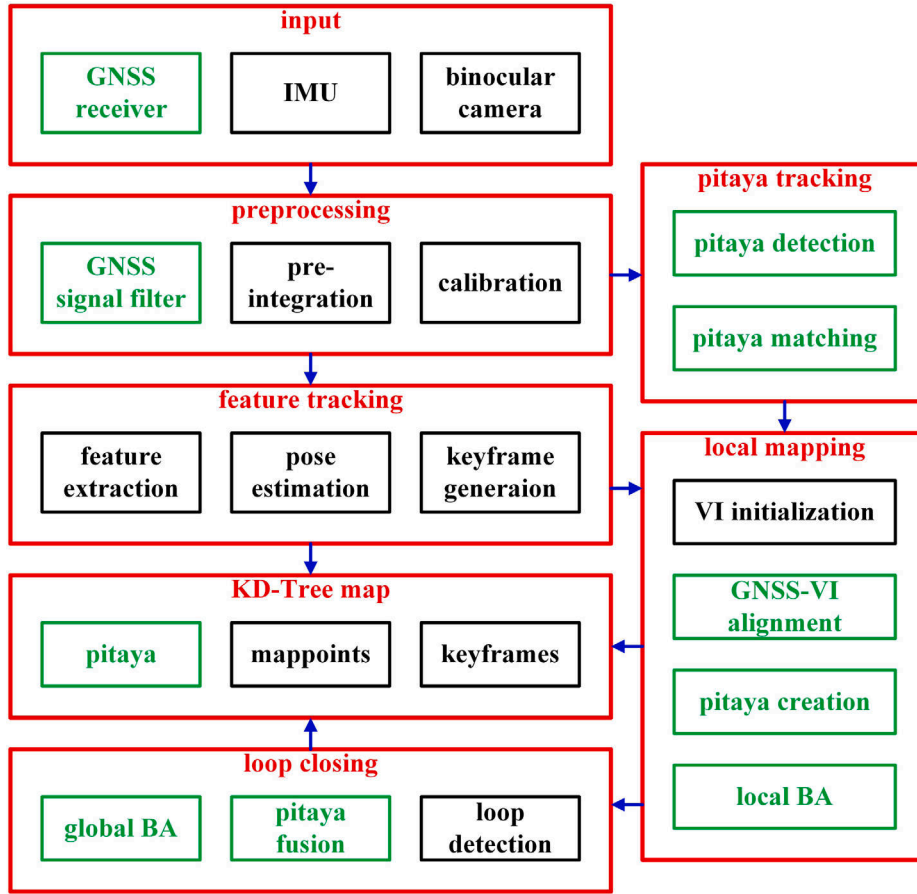


Fig. 2. System framework of SLAM-PYE. The main thread of the system is represented in red, the approaches related to GNSS raw measurement processing and pitaya state estimation are in green, and blue arrows indicate data flow.

illustrated in Fig. 1. In terms of software, the system framework consists of seven interconnected threads, depicted in Fig. 2. The preprocessing thread receives raw measurements from hardware-synchronized sensors, processes the data, and then transmits crucial information to both feature tracking and pitaya tracking, as detailed in Section 3.1. Section 3.5 details the pitaya tracking thread, which determines the 2D position of the pitaya in the image. Concurrently, the feature tracking thread establishes the initial position and orientation of the device for Bundle Adjustment (BA).

Considering the cost and stability, our system does not utilize hardware capable of directly measuring the distance, such as the depth camera, lidar, or robotic arm. Instead, we present an analytical algorithm in Sections 3.6 and 3.7 that calculates both the position and radius of the pitaya in a single-view measurement from the binocular camera. This algorithm provides the initial estimation for BA. Because of the variation in perspective and distance between the camera and the pitaya during movement, obtaining consistent state estimation across multi-view measurements poses a challenge when relying solely on analytical algorithms. We propose an approach that considers the pitaya state as variables and optimizes them through residuals from multi-view measurements. This approach is more effective in eliminating outliers and fusing pitaya from various perspectives, thereby leading to reliable and accurate estimation. Visual reprojection residuals on position and radius are also detailed in Sections 3.6 and 3.7.

To prevent the same pitaya from being repeatedly counted due to occlusion or multiple occurrences, our system integrates GNSS raw measurements into BA. These measurements are specifically divided into the pseudorange factor that constrains the device's position as detailed in Section 3.3, and Doppler factor that constrains the device's velocity as detailed in Section 3.4. However, since sensor data is

presented in various coordinate systems, it requires initialization and alignment before utilization. Consequently, in the local mapping thread of Fig. 2, there is a sequential execution of Vision-Only SFM, VI-Initialization, and GNSS-VI Alignment, all contingent on the device's motion. Section 3.2 describes the approach to aligning the local coordinate system post-VI-Initialization in three axes with the Earth-centered, Earth-fixed (ECEF) coordinate system of GNSS.

For real-time performance, the dense point clouds and textures calculated in SFM are ignored, and the system only estimates the device state $\mathbf{d}$, sparse mappoint position $\mathbf{m}$, and pitaya state $\mathbf{p}$. The pitaya state includes its position and radius, while the device state includes its position, orientation, and velocity. Furthermore, the system's BA assumes the independence of all measurements, with their noise following a zero-mean Gaussian distribution. Thus, similar to the derivation in Cao et al. (2022), the maximum a posteriori problem (MAP) involving the above state is reformulated to minimize the sum of residuals from multiple measurements, as depicted in Eq. (1). In this equation, $\mathbf{V}$, $\mathbf{I}$, and $\mathbf{G}$ represent the binocular image, inertial data, and GNSS raw measurements, respectively, while $r(\dots)$ represents the residuals on the specific state measured by these sensors. It is noteworthy that the Inertial Measurement Unit (IMU) and GNSS lack the capability to directly measure mappoint position and pitaya state. Therefore, these variables are not included in the GNSS and inertial residuals of the optimization problem.

$$\begin{aligned}
 (\mathbf{d}, \mathbf{p}, \mathbf{m})_{\text{MAP}}^* &= \arg \max p(\mathbf{d}, \mathbf{p}, \mathbf{m} \mid \mathbf{V}, \mathbf{I}, \mathbf{G}) \\
 &= \arg \min \{ \sum_{i=1}^m |r(\mathbf{V}_i, \mathbf{d}, \mathbf{p}, \mathbf{m})|^2 + \\
 &\quad \sum_{i=1}^n |r(\mathbf{I}_i, \mathbf{d})|^2 + \sum_{i=1}^o |r(\mathbf{G}_i, \mathbf{d})|^2 \}.
 \end{aligned} \tag{1}$$

Each residual in Eq. (1) is only related to several state (Kschischang et al., 2001), thus it can be transformed into the probability

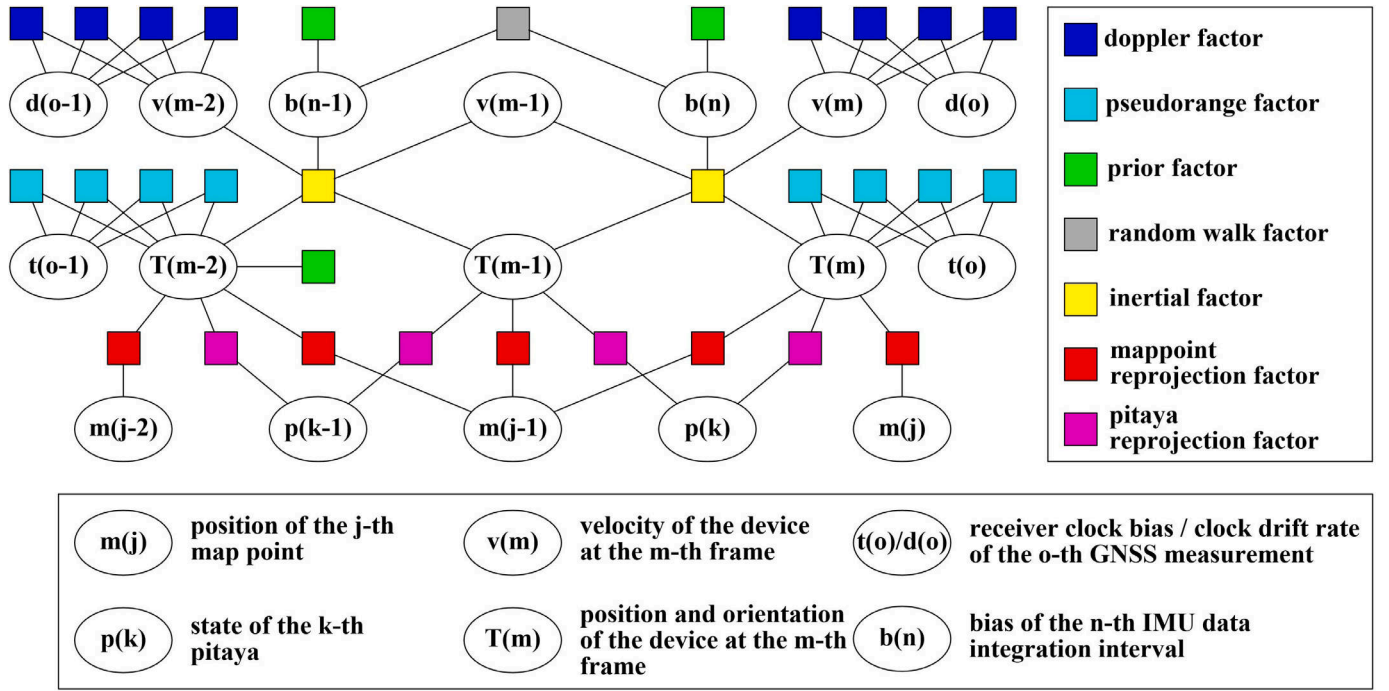


Fig. 3. Probability factor graph of the optimization problem after GNSS-VI alignment in our system, with a sliding window size of three. The annotated ellipse represents the state to be optimized in the system, such as the position and radius of pitaya. Colored rectangles represent the factors used to constrain the system's state, such as four pseudorange factors simultaneously constraining the position of the device. As the number of sliding windows increases, more factors and variables are involved in the optimization, resulting in a smoother estimated state with more computation time. Therefore, a real-time system needs to find a balance between computing resources and the number of sliding windows.

factor graph shown in Fig. 3. As depicted in Eq. (2), the GNSS raw measurement G is refined into the pseudorange factor P_{IFr}^s on device position T_d in Section 3.3, and the Doppler factor D_r^s on device velocity v_d in Section 3.4. In addition, the binocular image V is used not only to constrain the mappoint position m as introduced in Mur-Artal and Tardós (2017a), but also to constrain the pitaya state p . The reprojection residual on pitaya state $r(V, p)$ is refined into the residual on position $r(V, p_p)$ introduced in Section 3.6, and the residual on radius $r(V, r_p)$ introduced in Section 3.7. The IMU data I is refined into the random walk factor, ensuring the availability of the measurements, and the inertial factor, which constrain the movement between the two frames as introduced in Campos et al. (2021). For real-time purposes, the local BA optimizes only the state within the sliding window, with the other state serve as priori factors.

$$r(G, d) = r(P_{IFr}^s, T_d) + r(D_r^s, v_d), \quad (2a)$$

$$r(V, p) = \rho_{Hub}(r(V, p_p)) + \rho_{Hub}(r(V, r_p)). \quad (2b)$$

Incorrect matching and fusion of pitaya data can result in significant reprojection residual of the pitaya state, further leading to the estimated overall state deviating from the normal range and causing optimization failure. To address this issue, our system employs three strategies to identify and eliminate outliers. First, a robust kernel, such as the Huber cost function, $\rho_{Hub}(\dots)$, is used to reduce the weights of measurements with large residuals (Kümmerle et al., 2011), as shown in Eq. (2). Second, edges with residuals exceeding a threshold are removed from the factor graph. This threshold is affected by many factors, such as the image resolution, the confidence level of pitaya, and projection distance. The final strategy is based on manually defined attributes, such as the vertical and radius distributions of regular pitaya in Section 4.5.

3. Methodology

3.1. Preprocessing

Our system centers on frames captured by a binocular camera, utilizing IMU data and GNSS raw measurements to guarantee the smooth and consistent estimate of the pitaya's position and radius across long-range maps. However, due to the different measurement principles and output formats of these three sensors, we employ a preprocessing thread to extract the essential information.

Sensor calibration is an essential preparatory step prior to deploying a multi-sensor system. Our system entails two calibration steps: calibrating the binocular camera with the IMU, and calibrating the IMU with the GNSS receiver. Binocular matching relies on the assumption that identical features are projected at a consistent height in both the left and right images. Calibrating the binocular camera with IMU helps mitigate sensor and lens deviations that may arise during welding, assembly, and power-on, minimizing disparities between reality and the assumptions mentioned above. Calibrating between the binocular camera and the IMU is conducted using the excellent Kalibr toolbox, assuming a pinhole model camera and fixed temperature drift in the IMU (Rehder et al., 2016). Calibrating between the IMU and GNSS receiver facilitates the conversion between different coordinate systems. Given their proximity and the centimeter-level accuracy required, the distance between them is manually measured using a digital caliper. Since the IMU output frequency surpasses that of the camera and GNSS receiver, pre-integrating IMU data between frames reduces redundant calculations in the BA (Forster et al., 2017).

When processing GNSS raw measurements, the system initially excludes satellites with abnormal state registers, low signal-to-noise ratios, incomplete dual-frequency measurements, or low elevation angles to mitigate strong multipath interference in greenhouses. Subsequently, the terms of L_{MW} and L_{GF} , as presented in Eq. (3) and proposed by Blewitt (1990), are employed to combine the raw measurements and detect the cycle slip. Finally, the raw dual-frequency pseudorange are

Table 1
GNSS raw measurements used in the system.

GNSS system	Signal type	Frequency (MHz)	Pseudo-range	Doppler	Carrier phase
GPS	L1	1575.42	P_1	D_1	L_1
GPS	L2	1227.6	P_2	D_2	L_2
BDS	B1	1561.098	P_1	D_1	L_1
BDS	B3	1268.52	P_2	D_2	L_2

combined into an ionospheric-free pseudorange P_{IF} . In Eq. (3), L represents the carrier phase measurement, P represents the pseudorange measurement, and f_1 and f_2 denote the frequencies of two navigation signals within the same constellation, as detailed in Table 1.

$$L_{\text{MW}} = \frac{f_1 L_1 - f_2 L_2}{f_1 - f_2} - \frac{f_1 P_1 + f_2 P_2}{f_1 + f_2}, \quad (3a)$$

$$L_{\text{GF}} = L_1 - L_2, \quad (3b)$$

$$P_{\text{IF}} = \frac{f_1^2 P_1 - f_2^2 P_2}{f_1^2 - f_2^2}. \quad (3c)$$

3.2. GNSS-VI alignment

The GNSS receiver provides satellite positions, pseudorange and Doppler measurements in ECEF coordinate system, whereas the device's position, orientation, and velocity are described in the local coordinate system after VI initialization. Therefore, ensuring the global consistency of the map with the aid of raw GNSS measurements begins by aligning ECEF coordinate system with the local one. To facilitate the conversion between ECEF and local coordinates, an East-North-Up (ENU) coordinate system, centered on the device origin, is introduced into the system. In Eq. (4), the position p or velocity v in the local coordinate system $(\dots)^w$ is initially converted to the ENU coordinate $(\dots)^n$, and then to the ECEF coordinate $(\dots)^e$.

$$p_r^e = R_n^e R_w^n (p_r^w - p_{\text{anc}}^w) + p_{\text{anc}}^e, \quad (4a)$$

$$v_r^e = R_n^e R_w^n v_r^w. \quad (4b)$$

Before GNSS-VI alignment, there is insufficient information regarding the rotation R_w^n and translation p_{anc}^w that convert local coordinates to ENU coordinates, as well as the rotation R_n^e and translation p_{anc}^e converting ENU coordinates to ECEF coordinates. The position p_r^w and velocity v_r^w of the device in the local coordinate system, as well as the position p_r^e and velocity v_r^e of the GNSS receiver in ECEF coordinate system, are known. Therefore, GNSS-VI alignment involves adjusting rotation and translation to minimize pseudorange and Doppler residuals. Eq. (4) indicates that velocity conversion between two coordinate systems is solely linked to rotation, whereas position conversion is affected by both rotation and translation. Therefore, our system employs a three-step approach, progressing from coarse to fine adjustments, to achieve GNSS-VI alignment.

Step 1: Estimate the position of the receiver solely based on the constraints imposed by the pseudorange factor, excluding the visual and inertial factors. The coarse position within the greenhouse, affected by multipath interference, served as the initial anchor point for GNSS-VI alignment. Findings from greenhouse experiments reveal an anchor point drift of approximately ten meters, although it notably mitigates the nonlinearity resulting from the rotation between ECEF and ENU coordinate systems.

Step 2: Estimate the rotation between ECEF coordinate system and the local coordinate system, taking into account the constraint imposed by the Doppler factor. The initial rotation for R_n^e is computed using Eq. (5), where ϕ and λ are the latitude and longitude of the initial anchor point in Step 1. Concurrently, the initial rotation for R_w^n is set as the identity matrix. In contrast to the approach outlined in Cao et al. (2022), which focuses solely on estimating single-axis yaw, our system

computes the rotation matrix for all three axes. This necessity arises from aligning the gravity direction estimated from low-cost IMU data with that in the ENU coordinate system, which proves to be challenging in experiments where the device is kept in motion.

$$R_n^e = \begin{bmatrix} -\sin\lambda & -\sin\phi \cos\lambda & \cos\phi \cos\lambda \\ \cos\lambda & -\sin\phi \sin\lambda & \cos\phi \sin\lambda \\ 0 & -\cos\phi & \sin\phi \end{bmatrix}. \quad (5)$$

Step 3: Estimate the position of the anchor point and the rotation from ECEF coordinate system to the local coordinate system, considering the constraints of pseudorange and Doppler factor. The accuracy of Doppler measurements surpasses that of pseudoranges, especially when strong multipath interference is present (Kubo, 2009). As a result, the rotation after Step 3 differs slightly from that in Step 2, but a more accurate anchor point is obtained. Furthermore, as the position p_r^w and velocity v_r^w of the GNSS receiver in the local coordinate system differ from those of the IMU $(\dots)^b$, a transformation is necessary, as illustrated by Eq. (6). The rotation R_b^w is set as the identity matrix, and the translation r_b^r is manually measured during calibration.

$$p_r^w = R_b^w p_b^w + r_b^r, \quad (6a)$$

$$v_r^w = R_b^w v_b^w. \quad (6b)$$

After completing the above three steps, the optimized anchor point, serving as the origin within the ENU coordinate system, becomes the origin of the local coordinate system. The three axes of the local coordinate system are aligned with those of the ENU coordinate system. Following a coarse-to-fine GNSS-VI alignment, the GNSS raw measurements are integrated into factor graphs for both local and global BA, in the form of pseudorange and Doppler factors.

3.3. Pseudorange factor

Pseudorange measurement, a fundamental output of a GNSS receiver, depicts the geometric distance between the receiver r and the satellite s . Thus, when the satellite position is known, pseudorange factor can be employed to constrain the position of the receiver. However, it encompasses a range of errors in signal generation, propagation, and processing. These errors primarily involve receiver clock bias δt_r , satellite clock bias δt_s , ionospheric delay I_r^s , tropospheric delay T_r^s , and measurement error ϵ_r^s , modeled as a zero-mean Gaussian distribution, as depicted in Eq. (7). The satellite's position p_s^e and clock bias δt_s are determined from the broadcast navigation message. The velocity of light in a vacuum is denoted by c , and the tropospheric delay T_r^s is estimated using the Saastamoinen model (Saastamoinen, 1972). Due to the ionosphere-free combination for dual-frequency pseudorange measurements, the ionospheric delay is absorbed by the measurement error ϵ_r^s .

$$P_{\text{IF}r}^s = |p_s^e - p_r^e| + c (\delta t_r - \delta t_s) + I_r^s + T_r^s + \epsilon_r^s. \quad (7)$$

The time systems varies among different navigation constellation, and a minimum of five satellite signals are required to deliver complete position and velocity services for our dual-constellation system. Thanks to the abundant and evenly distributed signals from both GPS and BDS, our system can receive more than a dozen satellite signals, even

within a greenhouse covered with double-layer film. As a result, GNSS degradation is not considered.

$$r(P_{IF_r}^s, T_d) = |R_z(\omega_E t_{f_{rk}}^{sj}) p_{sj}^e - \hat{p}_{rk}^e| + c(\hat{\delta t}_{rk} - \delta t^{sj}) + T_{rk}^{sj} - P_{IF_r}^{sj}. \quad (8)$$

The device's position in BA is constrained by the pseudorange residuals, as illustrated in Eq. (8). The time $t_{f_{rk}}^{sj}$ required for the signal to travel from the j th satellite to the k th receiver position is not instantaneous. To compensate for the Earth's rotation during this interval, we refer to the axis R_z and angular velocity ω_E of the Earth's rotation. The receiver's estimated position $\hat{p}_{rk}^e$ is calculated from the device's estimated position $\hat{p}_{b_i}^w$ using Eqs. (6) and (4). The estimated pseudorange represents the geometric distance between the satellite's position p_{sj}^e and the receiver's estimated position $\hat{p}_{rk}^e$. The disparity between the estimated pseudorange and the actual pseudorange measurements, adjusted for the clock biases $\hat{\delta t}_{rk}$ and δt^{sj} of the receiver and satellite, as well as the tropospheric delay T_{rk}^{sj} , forms the pseudorange residuals $r(P_{IF_r}^s, T_d)$ in Eq. (2).

3.4. Doppler factor

It is well-known that when the relative velocity between the receiver and the satellite is non-zero, the frequency of the signal received by the receiver experiences a deviation from the fixed frequency set by the satellite. The greater the relative velocity, the more significant the deviation between the received signal frequency and the set frequency becomes. Thus, when the satellite velocity is known, Doppler factor can be employed to constrain the velocity of the receiver. The relationship among receiver velocity v_r^e , satellite velocity v_s^e , and Doppler measurements D_r^s is defined in Eq. (9).

$$D_r^s = -\frac{1}{\lambda} (\mathbf{1}_r^s (v_s^e - v_r^e) + \delta f_r - \delta f^s + \dot{f}_r^s + \dot{f}_r^s) + \epsilon_r^s. \quad (9)$$

In Eq. (9), δf_r represents the unknown drift in receiver clock frequency, δf^s denotes the drift in satellite clock frequency calculated from the broadcast navigation message, and $\mathbf{1}_r^s$ signifies the unit vector pointing from the receiver to the satellite in ECEF coordinate system. Considering the ionosphere and troposphere undergo slight changes over millisecond timescales, their rates of change, denoted as $\dot{f}_r^s$ and $\dot{f}_r^s$, are integrated into random errors ϵ_r^s . The random error is also modeled as a zero-mean Gaussian distribution.

$$r(D_r^s, v_d) = \frac{1}{\lambda} (\mathbf{1}_{rk}^{sj} (v_{sj}^e - v_{rk}^e) + \delta \hat{f}_{rk} - \delta f^{sj}) + D_{rk}^{sj}. \quad (10)$$

The Doppler factor constrains the device velocity in BA, as depicted in Eq. (10). The calculation of the Doppler residuals is similar to that of the pseudorange residuals in Section 3.3. The receiver's estimated velocity v_{rk}^e is calculated from the device's estimated velocity $\hat{v}_{b_i}^w$ using Eqs. (6) and (4). The estimated Doppler represents the difference between the satellite velocity v_{sj}^e and the estimated receiver velocity v_{rk}^e . The disparity between the estimated Doppler and the actual Doppler measurements, adjusted for the clock frequency drifts $\delta \hat{f}_{rk}$ and δf^{sj} of the receiver and satellite, is represented as the Doppler residuals $r(D_r^s, v_d)$ in Eq. (2).

3.5. Pitaya tracking

To simplify annotation and calculation, contrasting the 3D tomato tracking in point clouds by Rapado-Rincón et al. (2023), the pitaya tracking in our system is divided into two components: detection and tracking, both employing 2D real-time state-of-the-art image algorithms. The well-known YOLOv8 network is employed to extract the projected position and radius of the pitaya in the image (Jocher et al., 2023). To ensure robust performance in greenhouse experiments, we construct a dataset comprising 1267 images with varying sizes, illumination, and maturity levels for training the network. As shown in

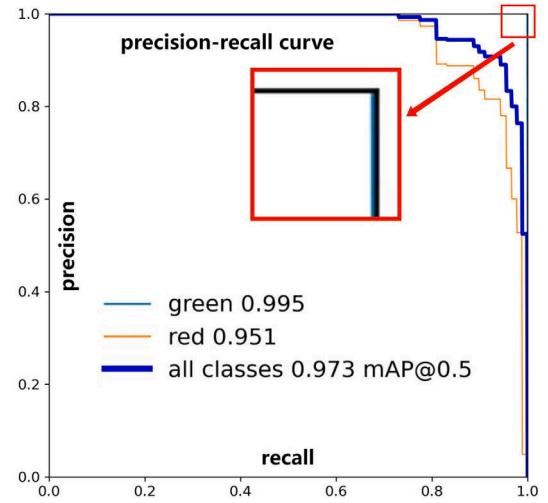


Fig. 4. The performance of the YOLOv8 network on the self-constructed pitaya dataset. Green and red represent immature and mature pitaya, respectively, and the system requires the position and quantity of the latter. The precision reaches one when the confidence level is below 0.834, indicating the absence of false positives in the system's results. The recall is greater than zero in all confidence levels, indicating that our system did not recognize pitaya in exceptional cases. Detailed cases and corresponding coping strategies are presented in Section 5.3.

Fig. 4, The Mean Average Precision (mAP@0.5) of the neural network on this dataset achieves an impressive 95.1%. Our system maintains the unchanged network architecture, and thus, the YOLOv8 network output includes the coordinates $[a \ b]^T$ of the upper-left corner, length c , width d , and the confidence level of the pitaya. The projected position $p_p^i : [x \ y \ z]^T$ and projected radius r_p^i of the pitaya are determined by Eq. (11). $(\dots)_l$ or $(\dots)_r$ indicates whether pitaya is in the left or right image.

$$p_p^i = \begin{bmatrix} a_l + 0.5 \times c_l \\ b_l + 0.5 \times d_l \\ a_r + 0.5 \times c_r \end{bmatrix}, \quad (11a)$$

$$r_p^i = 0.5 \times 0.7 \times \sqrt{c_l \times c_l + d_l \times d_l}. \quad (11b)$$

The tracking algorithm selected by our system is Deep OC-Sort (Maggiolino et al., 2023), which assigns a consistent ID to each pitaya across consecutive frames. It is crucial to emphasize that the deduplication process for pitaya in the tracking algorithm only works in adjacent frames. As depicted in Fig. 5(a), if a pitaya reappears after being obscured or moving out of the image, it will be assigned a new ID, even if it is the same pitaya in reality. Consequently, the system requires assistance from GNSS measurements to establish a globally consistent map, thereby preventing repetitive counting.

3.6. Estimation of pitaya position

Pitaya that appear in both the left and right images with a confidence level above the threshold, along with similar heights and radii, are treated as matching pairs. Once the projected position p_p^i of a matching pair is known, the position p_p^e of the pitaya in the 3D camera coordinate system can be determined using Eq. (12). Additionally, the position of the pitaya p_p^w in the local coordinate system is derived based on the camera's position t_c^w and orientation R_c^w . The intrinsic parameters of the camera include f_x , f_y , c_x , and c_y , whereas b represents the baseline of the binocular camera.

$$d = \frac{f_x \times b}{p_p^i \cdot x - p_p^i \cdot z}, \quad (12a)$$

Table 2
Sensors and equipment used in the experiments and their detailed specifications.

Sensor Type/Item	Value	Unit
Camera		
Sensor	AR0234	
Shutter	Global	
Resolution	1280 × 720	pixel
Horizontal field of view	100	degree
Vertical field of view	80	degree
Meas. frequency	20	Hz
IMU		
Sensor (Built-in)	BMI-088	
Meas. frequency	200	Hz
Gyroscope noise	1.89e−3	$\frac{\text{rad}}{\text{s} \times \sqrt{\text{Hz}}}$
Accelerometer noise	1.87e−2	$\frac{\text{m}}{\text{s}^2 \times \sqrt{\text{Hz}}}$
GNSS		
Receiver1	ZED-F9P	
Meas. frequency	10	Hz
RTK frequency	10	Hz
Receiver2	UM982	
Meas. frequency	20	Hz
RTK frequency	20	Hz
Prime 17W		
View number	10	
Range	6 × 6 × 3	m
Solution frequency	200	Hz
Precision	0.1	mm

Table 3
The third-party open-source libraries used in main approaches of the system.

Library	Function	Version	Reference
RTKLIB	RINEX decoding	2.4.3	Takasu et al. (2009)
OpenCV	image processing	3.4.18	Bradski (2000)
g2o	graph optimization	1.0.0	Kümmerle et al. (2011)

approximation of pitaya, our system uses a cubic polynomial with respect to the radius, as detailed in Eq. (16), to fit the weight. Using the least squares algorithm, we calculate the three parameters $a = 0.011$, $b = -0.521$, and $c = 14.656$ from 54 pairs of weights and radii, measured using the electronic scale and digital caliper depicted in Fig. 1. Because of measurement and assumption errors, the coefficient of determination R^2 for the fitted equation is 0.852.

$$w_p = a \times r_p^{w3} + b \times r_p^{w2} + c \times r_p^{w1}. \quad (16)$$

4. Experiment

4.1. Experimental setup

As depicted in Fig. 1, our system is deployed on a handheld platform that includes GNSS receivers, an IMU, and a binocular camera. The detailed hardware specifications are shown in Table 2. Besides, we utilize several third-party open-source libraries, as listed in Table 3, to promptly verify the system's feasibility. To tackle the inconsistent trigger frequency of the these sensors, the platform synchronizes the timestamps of the camera and IMU with Universal Time Coordinated (UTC) by leveraging the Pulse Per Second (PPS) signal of the UM982. All experiments in this section are conducted on a desktop computer featuring an Intel i7-8700 CPU clocked at 3.8 GHz, 32 GB of RAM, and an NVIDIA RTX-3060 GPU.

Indoor experiments are conducted at the Information Science Experiment Center of USTC, as illustrated in Fig. 7. Because of the weak indoor GNSS signal, we verify the accuracy of the pitaya's position estimated by our system using the optical 3D positioning device, Prime 17 W. We collect four sequences (A, B, C, and D) with references, adjusting the distance of the trajectory from the shelves while maintaining the pitaya's position and radius constant.

The greenhouse experiments are conducted at a commercial pitaya plantation situated on the outskirts of Hefei City, Anhui Province. The pitaya species cultivated in the plantation is *Hylocereus costaricensis*, which is easily distinguished by its red fruits against the green branches. An individual mature greenhouse is selected to validate our system's yield estimate, as depicted in Fig. 8(b). The experiments are conducted in December 2023, with the greenhouse being covered by a double-layer plastic film to retain warmth. Prior to the experiments, twenty pitaya on the branches are selected and numbered. The sampled pitaya are fresh, ripe and free from rot, with a radius within the normal range, suitable for marketing. In addition, in order to verify the accuracy of the system's estimates over long distances and to facilitate manual confirmation of the system's results, the spacing between sampling points is 2–5 m. The distances between them are measured using a tape measure, serving as a reference for positioning. After collecting two sequences (E and F), their actual radii and weights are measured using an electronic scale and a digital caliper as references.

Our system employs the evaluation metrics outlined by Miranda et al. (2023a), as shown in Eq. (17). The Root Mean Square Error (RMSE) quantifies the position and radius error, while the Mean Absolute Percentage Error (MAPE) quantifies the weight error. In addition, the Coefficient of Variation (CV) quantifies the stability of the system in estimating weight within the same sequence. x_i denotes the reference value, $\hat{x}_i$ denotes the estimated value, $\bar{x}$ denotes the mean of the estimated values, and σ denotes the sample standard deviation of the estimated values. n denotes the number of samples, with 24 samples for indoor experiments, 20 for greenhouse experiments, and 5 for estimates number of each sequence.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (\hat{x}_i - x_i)^2}{n}}, \quad (17a)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{x}_i - x_i}{x_i} \right| \times 100\%, \quad (17b)$$

$$\text{CV} = \frac{\sigma}{\bar{x}} \times 100\%, \quad (17c)$$

$$\text{where } \sigma = \sqrt{\frac{\sum_{i=1}^n (\hat{x}_i - \bar{x})^2}{n-1}}. \quad (17d)$$

4.2. Position accuracy analysis

To mitigate the impact of randomness introduced during feature selection in feature tracking thread, Table 4 presents the RMSE between 30 estimates generated by our system from six sequences and their corresponding references. Fig. 9 is a visualization of a single estimate from indoor experiments, showing a map containing device's motion trajectory and 24 simulated pitaya models.

When comparing the horizontal and vertical position accuracy, disparities emerge between indoor and greenhouse experiments. In indoor experiments, the horizontal position estimation error is around 20 mm, surpassing the error in the vertical direction. In greenhouse experiments, both horizontal and vertical accuracies are approximately 50 mm, indicating minimal disparity. This discrepancy may stem from the ellipsoidal pitaya's transverse positioning in indoor experiments, approximated as a sphere, leading to uncertainty in the x -axis coordinate within the image coordinate system output by the pitaya tracking thread. While the orientation of the pitaya in the greenhouse is arbitrary.

Regarding the comparison of positional error in the form of Euclidean distance between the indoor and greenhouse experiments, the former achieves approximately 30 mm, while the latter reaches about 70 mm. The positioning of the pitaya relies on the device's coordinates. As a result, sequences E and F, derived from long-range greenhouse experiments, manifest higher pitaya positional errors in contrast to those observed in indoor experiments with motion restrictions.

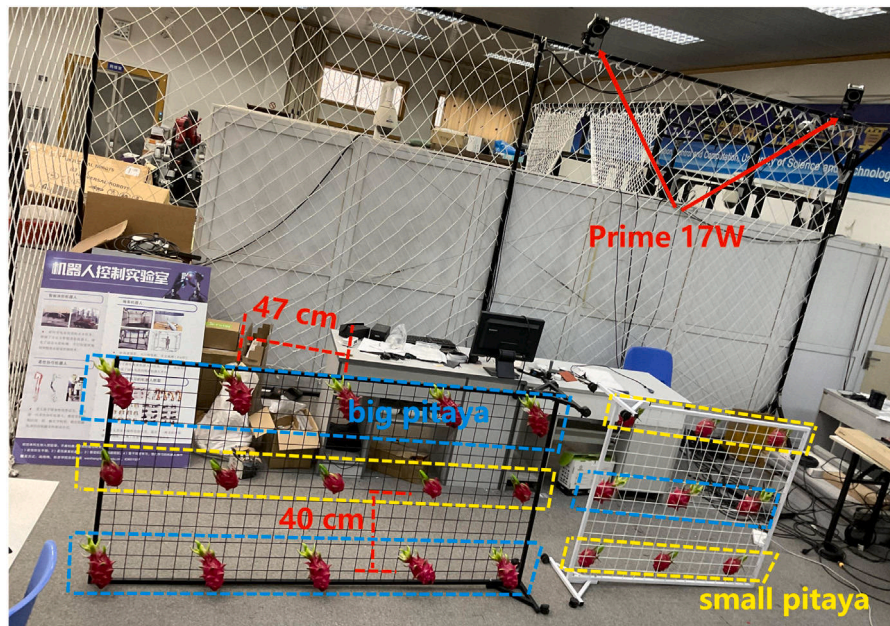


Fig. 7. In indoor experiments, the simulated pitaya models are fixed on shelves with uniform spacing. The pitaya's radius is larger in the blue area than in the yellow area. A digital caliper is utilized to measure the reference radius in indoor experiments.

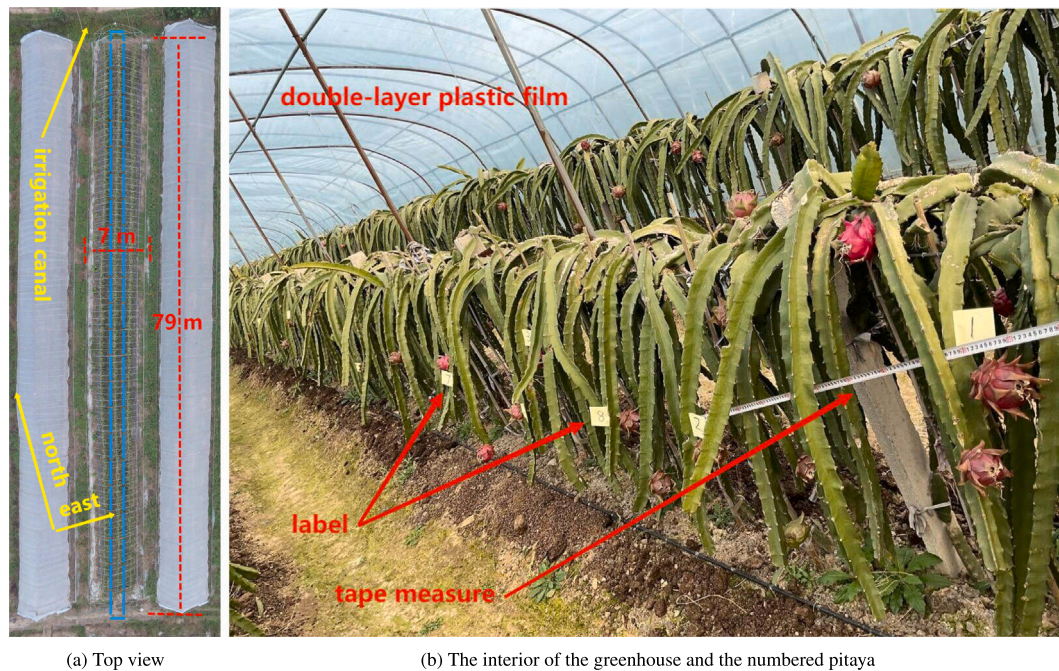


Fig. 8. Commercial pitaya plantation. The left image, captured in October, illustrates an outdoor temperature of approximately 20 °C, with the pitaya plantation left uncovered by polyethylene plastic film. The red line marks the dimensions of the greenhouse, the blue line indicates the experimental area, and the yellow line indicates that the northeastern part of the greenhouse is adjacent to the irrigation canal. The right image, captured in December when the outdoor temperature was approximately 10 °C, depicts the greenhouse covered with a double-layer film, thus maintaining an indoor temperature of around 25 °C. Twenty pitaya are numbered, and distances between them are measured using a tape measure as references.

Moreover, to assess the correlation between projection distance and accuracy, we intentionally increased the gap between the trajectory and two shelves incrementally from sequence A to sequence D during indoor experiments. With an increase in z as indicated in Eq. (13), x necessitates a larger positional correction than z to yield an equivalent residual. Consequently, the positional error exhibits a gradual increase from sequence A to sequence D.

4.3. Radius accuracy analysis

Accurate pitaya detection and decoupling the pitaya radius estimation from the absolute position of the device contribute to a smaller error in radius estimation compared to positional error. In indoor experiments detailed in Table 4, the median radius RMSE for the large pitaya is 6.4 mm, while for the small pitaya, it is 5.6 mm. The large size

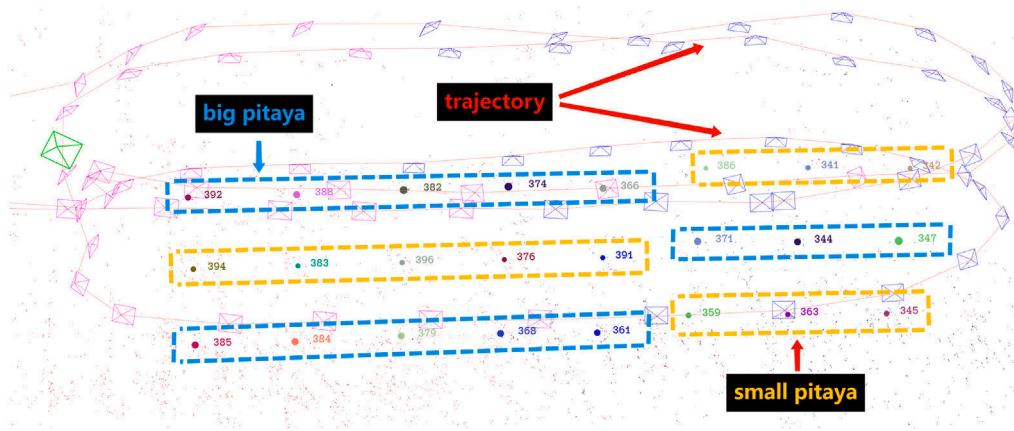


Fig. 9. A map containing pitaya created in indoor experiments. Each colored sphere represents an individual pitaya, and corresponds to the simulated pitaya model illustrated in Fig. 7. The red line illustrates the trajectory of the device's movement. All pitaya on both shelves are created, and their radii are accurately estimated.

Table 4

RMSEs for the estimated position and radius of the pitaya. Six sequences, denoted as A, B, C, D, E, and F, are collected during the experiments. The first four sequences are acquired indoors, while the last two are acquired in the greenhouse. As there are no large and small simulated pitaya models in greenhouse experiments, the corresponding entries are left blank. Results for five typical estimates are presented for each sequence, so there are 30 entries in total in the table.

Sequence	Horizontal RMSE (mm)	Vertical RMSE (mm)	Large radius RMSE (mm)	Small radius RMSE (mm)
A1	19.5	12.3	6.8	4.9
A2	20.8	11.8	5.0	2.9
A3	19.9	12.1	5.9	2.5
A4	18.7	19.1	8.3	3.3
A5	20.5	12.1	8.5	3.6
B1	22.1	16.6	5.6	5.8
B2	22.8	17.9	8.1	6.6
B3	23.6	17.0	9.2	6.4
B4	21.3	18.1	8.6	6.4
B5	21.9	22.2	3.8	3.9
C1	23.3	19.0	4.8	5.6
C2	22.0	20.6	5.0	5.6
C3	24.3	22.0	7.0	8.1
C4	23.6	18.6	8.6	8.1
C5	19.6	17.3	9.8	7.6
D1	22.1	18.1	7.0	4.8
D2	22.5	19.9	4.6	5.2
D3	21.8	19.6	4.8	5.7
D4	31.8	26.8	5.4	6.5
D5	23.5	20.0	4.5	5.6
E1	47.6	39.0		
E2	47.9	61.1		
E3	61.5	53.1		
E4	51.5	43.1		
E5	39.3	45.0		
F1	46.3	54.0		
F2	42.8	51.1		
F3	51.0	49.6		
F4	45.5	57.1		
F5	47.1	45.1		

of the pitaya leads to less accurate estimation of m and n in Eq. (14), resulting in lower accuracy in estimating the radius compared to the small one.

Table 5 displays the system's radius estimates for two sequences conducted in the greenhouse, along with their respective errors. In greenhouse experiments, the median radius RMSE is 7.1 mm, which is slightly higher than that in indoor experiments. Possible reasons include an inadequate number of measurements resulting from rapid motion and occlusion due to complex environments. Nevertheless, these errors remain below the actual pitaya radius of approximately 40 mm, thereby confirming the feasibility of radius estimation by our system.

To uphold the precision of the system's yield estimation, pitaya located at distances exceeding twenty times the baseline are ignored,

thereby diminishing the clarity of the relationship between radius error and projection distance.

4.4. Yield accuracy analysis

Following the exclusion of pitaya with unreasonable positions and radii in the map, the total weight derived from summing the individual weights calculated from the radii represents the estimated yield of a single greenhouse. While it is feasible to harvest all pitaya from the entire greenhouse and weigh them to verify accuracy, such an action is wasteful. Consequently, we assess accuracy based on the twenty sampled pitaya. All twenty pitaya in our system have been estimated and none have been identified as outliers. Their mean weights and MAPEs are listed in Table 5.

Table 5

Errors in the radius and weight of pitaya in greenhouse experiments.

Sequence	Mean radius (mm)	Radius RMSE (mm)	Mean weight (g)	Weight MAPE (%)
E1	40.3	8.6	471.8	17.4
E2	39.7	5.1	459.9	13.9
E3	39.6	8.2	455.6	15.7
E4	39.6	7.1	454.3	14.9
E5	38.3	6.9	422.4	21.5
F1	40.5	6.0	492.4	18.2
F2	38.5	5.7	445.7	16.0
F3	40.2	8.6	433.7	17.1
F4	39.4	7.1	464.5	15.3
F5	40.2	8.1	485.9	13.2

Table 6

The time consumption of the main approaches in our system. Given that the speed of the feature tracking thread (feature extraction + pose estimation) surpasses the camera output speed, our system runs in real-time.

Sequence	Pitaya detection (ms)	Feature extraction (ms)	Pose estimation (ms)	Pitaya creation (ms)	Local BA (ms)
A	12.6	20.4	6.6	113.7	256.7
B	13.1	23.8	10.7	94.9	293.4
C	13.5	22.7	7.2	112.2	307.2
D	12.9	22.7	8.4	115.1	222.2
E	12.5	21.7	7.6	38.8	145.0
F	12.9	22.2	12.1	37.6	108.1

According to Eq. (17), the CV of the estimated mean weight for the sequence E is 4.1%, while that for the sequence F is 5.5%, which indicates the stability of our system. Additionally, most of the weight MAPEs in Table 5 remain within 20%, with a median of 15.7% in sequence E and 16% in sequence F. We manually checked and confirmed that the system does not include repetitive counting in both sequences. Consequently, based on the median weight MAPE of 20 pitaya sampled from the 97 pitaya created in sequence E, we speculate that the yield estimation error of our system for the entire greenhouse is approximately 16%. Various factors will influence this result, notably the significant deviation in the shape of the pitaya from the ellipsoid assumed by the system. Additionally, the system encounters the challenge of missing counting, as discussed in Section 5.3.

4.5. Distribution analysis

In addition, each pitaya created by our system is assigned a unique position, which facilitates analysis of pitaya distribution rather than just a total yield. The horizontal distribution portrayed in Fig. 10 indicates that pitaya are primarily concentrated in the northeastern part of the greenhouse, with fewer occurrences in the central and southwestern parts. Given that the northeastern part of the greenhouse depicted in Fig. 8(a) is adjacent to an irrigation canal, providing more abundant water and nutrients, this factor might have contributed to this distribution pattern. Regarding vertical distribution, the growth of pitaya shows no noticeable preference for height. They attach themselves to branches that are secured to shelves, ensuring that the height of the pitaya remains below that of the shelves. The height being less than one meter not only facilitates manual harvesting but also allows a camera-equipped device to observe most pitaya without requiring vertical movement. The radius distribution is also generally uniform, with the majority measuring approximately 40 mm. Therefore, the system leverages these regular vertical and radius distributions to eliminate outliers.

4.6. Time analysis

Table 6 presents the time consumption of the main approaches across various sequences in our system. Pitaya detection takes approximately 12 ms, employed to forward propagate the image through the neural network, and remains constant regardless of the number of pitaya. The time consumption of feature extraction remains relatively

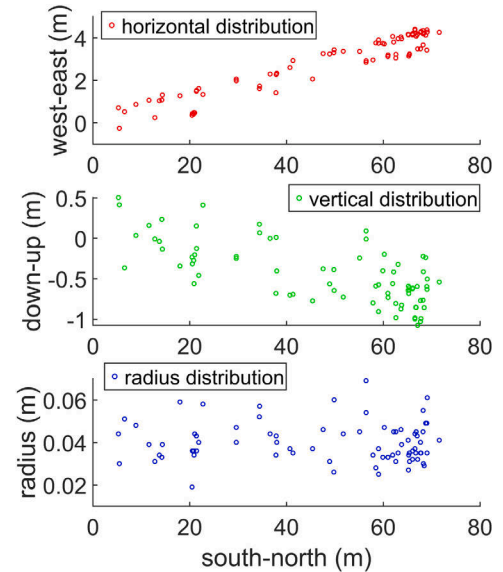


Fig. 10. Distribution of pitaya positions and radii estimated in greenhouse experiments, and each circle represents a pitaya. The x-axis represents the relative position in the south–north direction, and the y-axis represents the relative values in the west–east direction, vertical direction, and pitaya radius, respectively. Because the greenhouse is not strictly aligned with the ENU coordinate system, the circles in the upper figure are distributed on a slope. To facilitate drainage during the rainy season, the greenhouse is designed with higher elevation in the southwest and lower elevation in the northeast, resulting in the tilted representation of relative altitude in the middle figure.

stable, as the parameters are fixed at 800 feature points and 5 layers of the pyramid. If there is no pitaya in the sliding window, the pitaya creation in Fig. 2 is skipped, so its time consumption is dependent on the number of pitaya in the window. Owing to the lower density of pitaya in the greenhouse compared to indoors, sequences E and F require less time for pitaya creation.

Time consumed by local BA is similar to pitaya creation and correlates with the number of state and residuals. Although pseudorange and Doppler residuals in greenhouse experiments can constrain the device's position, their number, being less than a hundred, has a minimal impact on time consumption. Indoor experiments involve numerous

Table 7

Hardware cost estimates for several typical orchard reconstruction and fruit positioning systems. In our system, the depth camera only outputs inertial data, so only the cost of the sensor chips is considered.

Reference	Robotic arm	Lidar	Depth camera	Binocular camera	IMU	GNSS receiver	Approximate prize (Dollar)			
Kang et al. (2022)	6-DOF	Livox Mid-70	Realsense D455	ZED2	BMI-088	UM982	1300			
Chen et al. (2021)		Livox Mid-70	Realsense D435 Azure Kinect Kinect v2				1200			
Liu et al. (2023)							1100			
Miranda et al. (2023a)			AR0234				900			
Gong et al. (2022)							200			
Ours							200			

overlapping perspectives, leading to a higher number of reprojection residuals. Due to the reduced number of residuals in greenhouse experiments with the same sliding window size, which is about half compared to the indoor one, sequences E and F exhibit faster processing than the first four sequences.

The bottleneck in processing speed within this system occurs within the feature tracking thread as depicted in Fig. 2. This thread integrates the Visual-Inertial Odometry (VIO) system, primarily involving feature extraction and pose estimation. The former operates at a speed of around 22 milliseconds per frame, while pose estimation takes approximately 10 milliseconds per frame. The average processing speed of a single VIO system is approximately 25 frames per second. Thanks to the multi-threaded framework, the overall system operates at an average speed of 23 frames per second. Since this speed exceeds the camera's output speed of 20 frames per second, our system ensures real-time performance.

4.7. Cost analysis

An affordable and easy-to-maintain device is appealing for farming with narrow profit margins. Table 7 illustrates the hardware costs for several typical orchard studies. Given that most pitaya are under one meter in height, our system does not necessitate a robotic arm for vertical motion. Lidar offers numerous advantages, and its price is expected to decrease with widespread adoption. Depth cameras are widely used in orchard studies for several reasons, including being cost-effective and easy to maintain, and offering denser depth and wider field of view than lidar at appropriate distances, reflections, lighting, and textures. (Fu et al., 2020). We also attempted to utilize depth camera in our system. In winter greenhouse experiments, we observed accurate depth estimation. However, under direct autumn sunlight, certain pixels exhibit inaccurate or non-existent depth readings, as encountered by Gong et al. (2022) or Kang et al. (2022). This challenge is caused by the measurement principle based on structured light or Time of Flight (ToF), and cannot be avoided. Therefore, we blocked the image and depth output of the depth camera. Our system does not use standard commodity kits, but rather abundantly available sensor chips, so there is room for further price reduction through integration.

5. Discussion

5.1. GNSS validity

A tightly coupled VIO system can achieve centimeter-level accuracy in small-range scenarios (Campos et al., 2021). Without global position and velocity constraints, a system utilizing low-cost sensors becomes insensitive to rotation, leading to distortion in the constructed map. Fig. 11(a) illustrates the erroneous creation of the same pitaya multiple times at different positions in the map, resulting in unreliable yield estimation. Consequently, our system integrates GNSS raw measurements to impose constraints on the device's position and velocity, ensuring consistency in the position and radius of the same pitaya across multiple measurements, as depicted in Fig. 11(b).

5.2. Multi-view measurement

Pitaya positions calculated analytically from single-view measurements often deviate from predefined distribution rules, posing challenges in achieving consistent estimates using methods like clustering. As illustrated in Fig. 12, variations in perspective and distance result in positional offsets, particularly in the depth direction, which can be ten times the pitaya's radius. Moreover, the occlusion by branches limits the measurement to a specific perspective, resulting in fixed uncertainty in that direction. Similar to estimating the pitaya position, measuring the pitaya radius from different perspectives and distances produces varying noise distributions, albeit significantly smaller than the noise distribution of the position. Hence, our system utilizes a unified reprojection residual to constrain both the position and radius of the pitaya. Measurements with excessive residuals are excluded as outliers to avoid interference with state estimation. The RMSE for the horizontal and vertical positions of the pitaya, calculated using the analytical algorithm from single-view measurements, is 134 and 237 mm, whereas our system achieves an RMSE within 30 mm in indoor experiments.

5.3. Misidentification

The complex greenhouse environment leads to unpredictable misidentifications by the pitaya detector. Increasing the size of the dataset or enhancing its distribution can reduce false positives. Expanding the dataset's size ensures that objects like the red bricks shown in Fig. 13(a) are no longer misclassified as pitaya. The manually discarded rotten pitaya depicted in Fig. 13(b) are filtered based on their heights above the ground. The dataset is segmented into mature and immature categories to better handle cases like the one in Fig. 13(c). Therefore, in both indoor and greenhouse experiments, once our system identifies an object as a pitaya in the map, it is guaranteed to be a pitaya in reality. In other words, there are zero false positives.

More challenging than false positives are false negatives, which reduce the estimated yield and pose difficulties in appropriate handling. Fig. 14(a) illustrates a case of treating multiple adjacent pitaya as one, a common phenomenon during concentrated maturation. A radius threshold is set by our system to suppress this error. The occlusion depicted in Fig. 14(b) minimally interferes with the multi-view measurement system since most pitaya can be fully exposed in certain perspectives. The system disregards the small pitaya depicted in Fig. 14(c). On one hand, this prevents the detector from accumulating sufficient features to make accurate judgments; On the other hand, in reality, it cannot be sold in the market.

In the entire sequence E, the detector detected a total of 294 pitaya in 2D images, and the system created 97 pitaya in the 3D map, out of which 19 are filtered out. The reasons for system filtering include cases where the radius or 3D position falls outside a reasonable range, the number of multi-view measurements is below the threshold, or the distance from the camera is unreasonable. The corresponding filtered numbers for these reasons are 5, 12, and 8, respectively. Some pitaya are filtered out due to multiple reasons simultaneously. Nonetheless, biases are introduced into the system and several normal pitaya is eliminated. The manual check results indicated that only 13 out of the 19 pitaya should have been filtered out, suggesting the need for further investigation into missing counting.

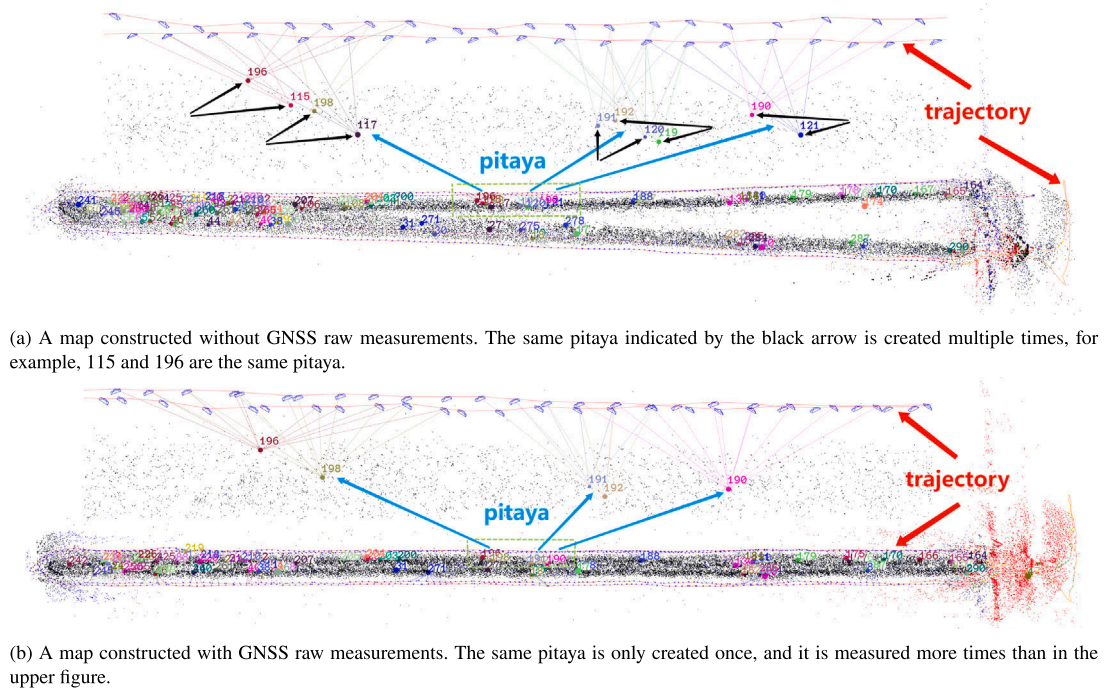


Fig. 11. The mappoints are represented by black dots, the device's motion trajectory is indicated by a red line, and pitaya is depicted as colored spheres. Each pitaya is measured multiple times, forming multiple colored lines connecting it to the trajectory. In the upper figure, the system exhibits low sensitivity to rotation, causing deviations between the trajectory's starting and ending points, with non-parallel back-and-forth trajectory. In the lower figure, the trajectory forms a closed loop under the constraints of the raw GNSS measurements. In addition, because the same pitaya in Section 3.5 is assigned multiple IDs and is created at different positions in the map, this makes yield estimation unreliable. In the lower figure, the pitaya fusion in Fig. 2, along with reprojection residuals from multi-view measurements, ensures a globally consistent estimate of both position and radius.

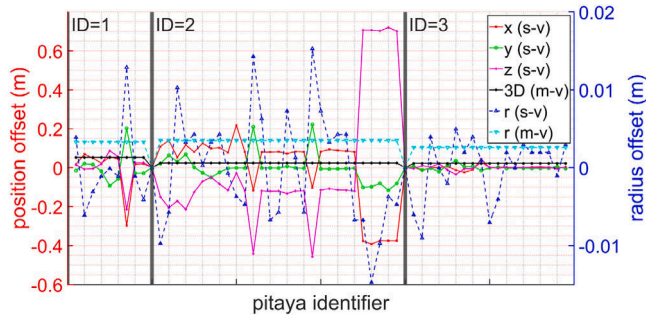


Fig. 12. Position and radius offset of three pitaya (ID = 1,2,3) estimated by two different algorithms. The x-axis corresponds to multiple estimates of three pitaya, while the y-axis represents the offset of the results. The original estimates are adjusted by subtracting the averages, ensuring they are centered at zero. The z-axis (depth direction) exhibits higher uncertainty in position offsets calculated from single-view measurements (s-v) by analytical algorithms. In multi-view measurements (m-v), the 3D position constrained by multiple reprojection residuals is more stable and accurate compared to the former, thanks to the outlier exclusion. The radius offset estimated by both approaches is roughly one order of magnitude smaller than the position offset.

6. Conclusions and future work

This article introduces a system that tightly couples complementary sensors for estimating the position and radius of pitaya, further estimating greenhouse yield. The system integrates raw measurements from a binocular camera, IMU, and GNSS receiver within the graph optimization framework, which provides a cost-effective and low-maintenance alternative compared to lidar or robotic arm. Only the state of sparse mappoints and pitaya are estimated, combined with inertial assisted visual navigation, enables the system to achieve real-time performance. The integration of dual-frequency GNSS raw measurements ensures the global consistency of the constructed map, even in a commercial greenhouse with strong multipath interference, while also avoids repetitive

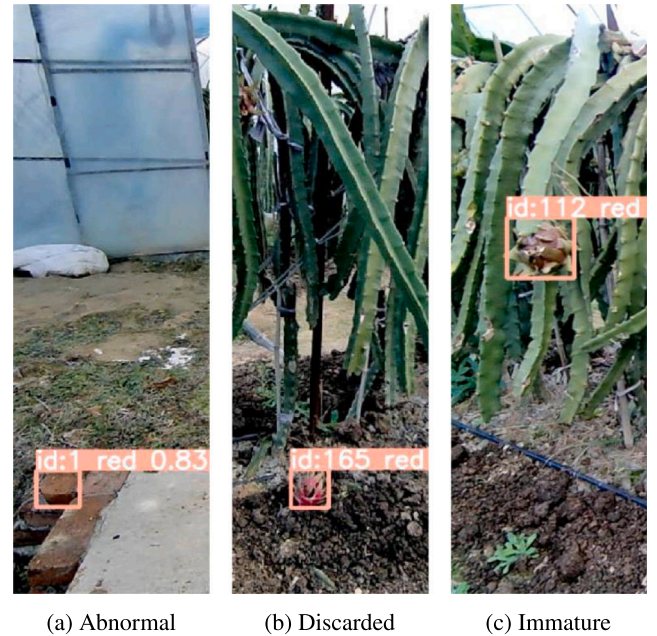


Fig. 13. Three cases of false positives. In Fig. 13(a), the detector mistakenly identifies a brick as a pitaya. In Fig. 13(b), the detector incorrectly re-identifies a discarded pitaya with wormholes on the ground. Fig. 13(c) shows a case of immature pitaya being misidentified.

counting. The pitaya state is constrained by the reprojection residuals from multi-view measurements in a unified probability factor graph, thereby facilitating outlier exclusion and improving radius estimation accuracy. Results from indoor and greenhouse experiments demonstrate that the position RMSE of pitaya is 7 cm, the radius RMSE is

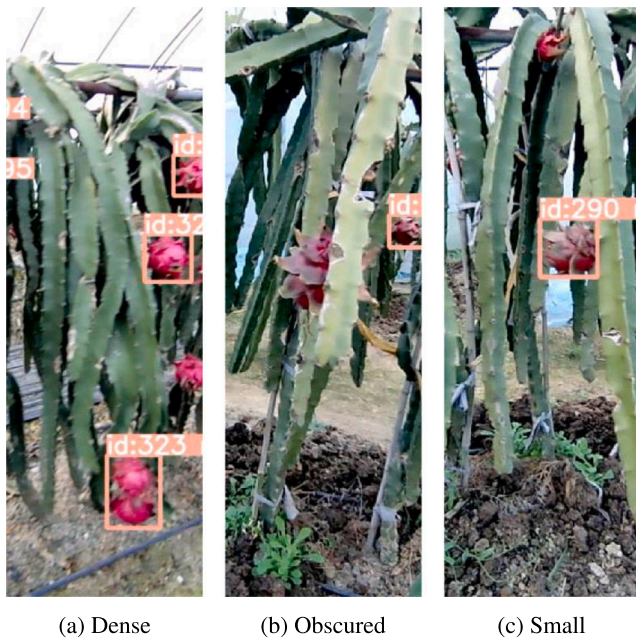


Fig. 14. Three cases of false negatives. In Fig. 14(a), the detector erroneously interprets the two closely positioned pitaya at the bottom as a single entity. In Fig. 14(b), the pitaya remains undetected due to obstruction by branches. In Fig. 14(c), despite being red, the upper pitaya is not recognized due to its premature ripeness and small size.

7 mm, and the weight MAPE of individual pitaya is about 16%. In conclusion, our system has demonstrated the feasibility of achieving real-time and accurate pitaya yield estimation through the integration of cost-effective sensors with general computing resources.

In greenhouse experiments, the vertical position accuracy of our system is lower than that of the horizontal one, which is consistent with the characteristics of pure GNSS measurements. Satellites are only distributed around and above, and their signals cannot penetrate the earth to propagate. As a result, there are more constraints in the horizontal direction than the vertical direction in the factor graph. Moreover, our pitaya tracking algorithm, originally designed for general objects, may result in missing counting for pitaya. For future research, we plan to explore and integrate additional low-cost sensors, such as barometers and magnetometers, into our systems to enhance the vertical accuracy. Additionally, we will improve the neural network architecture and try more challenging fruits, such as occluded and similar-colored pears, tall and dense apples, as well as low and small strawberries.

CRediT authorship contribution statement

Hongjie Wang: Writing – original draft, Software, Methodology, Conceptualization. **Xiangyu Hong:** Software, Data curation. **Linlin Qin:** Writing – review & editing, Supervision, Methodology. **Chun Shi:** Supervision. **Gang Wu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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